

Project Design Phase-II: Data Flow Diagram & User Stories

Date	2026-09-28
Team ID	SWTID-2026-4282
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. Data Flow Diagram (DFD) Levels & System Information Flow

A Data Flow Diagram (DFD) mapping visually represents how procurement data routes through ServiceNow Flow Designer. Below is the logical step-by-step information migration model for the automated laptop provisioning environment:

- 1. Intake Processing (Trigger):** The End-User configures a device request inside the Service Portal Catalog Item.
- 2. Profile Enrichment:** The system parses user session parameters via `gs.getUser()` to automatically fetch department, cost center, and corporate location constraints.
- 3. Inventory Scan (CMDB Data Store):** Flow Designer executes a background lookup block on the Asset Management Table (`alm_hardware`) to scan model availability.
- 4. Dynamic Approvals (Process Block):** The data engine routes structured approval payload pills directly to the User's Line Manager via MS Teams and Email integrations.
- 5. Fulfillment & Automated Provisioning:** The IntegrationHub REST API invokes enterprise cloud deployment engines (Windows Autopilot) to pre-stage software payloads based on user persona data, modifying the asset status directly inside the repository database from *Available* to *In Use*.

2. User Stories Reference Matrix

Use the structured table layout below to map all project deliverables, acceptance bounds, execution timelines, and lifecycle epics.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee (End-User)	Portal Request Intake	USN-01	As an employee, I can open the IT Portal and choose a standard laptop so that I don't select incorrect configurations.	• Catalog displays pre-approved laptop types based on user role. • Form auto-fills department data.	High	Sprint-1
	Order Visibility	USN-02	As an employee, I want to receive real-time updates regarding my order status so that I know when my laptop ships.	• Ticket dashboard displays live fulfillment tracking. • System auto-triggers email confirmations.	Medium	Sprint-1
	Device Authentication	USN-03	As an employee, I want to sign	• First login triggers Windows	High	Sprint-2

			in directly to my new device using corporate Azure AD credentials without manual setup.	Autopilot / ABM profile. • Core company apps install instantly.		
Line Manager (Approver)	Automated Approval Gate	USN-04	As a manager, I want to view and approve laptop orders straight from MS Teams or Email to prevent system delivery delays.	• One-click "Approve / Reject" option inside alerts. • SLA countdown alerts route back to the system.	High	Sprint-1
IT Asset Manager	Automated Inventory Control	USN-05	As an asset manager, I want the system to automatically update stock quantities in the CMDB when an item is assigned.	• Asset registry moves to "In Use" on delivery. • Serial numbers map automatically to the user profile.	High	Sprint-1

	Vendor Management	USN-06	As an asset manager, I want Flow Designer to auto-draft a purchase order to the vendor if inventory counts drop low.	• Trigger launches PO webhook when stock dips below 10 units. • Vendor receives structured hardware specs data.	Medium	Sprint-2
System Administrator	Error Handling & Auditing	USN-07	As a system administrator, I want to trace failed approval exceptions so that I can re-route stuck workflows.	• Flow contexts capture failed task state variables. • System triggers a fallback incident to the IT desk.	Low	Sprint-2